

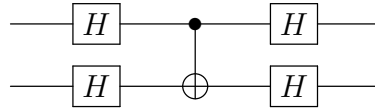
QUANTUM COMPUTATION - EXERCISE 2

Deadline: 14/05/2026 before midnight.

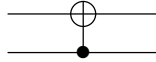
Submission **in triplets or quadruplets only**). Please upload your solutions to Moodle. We expect nothing less than fully detailed answers. Questions, extension requests, or any other issue related to the exercise should be emailed to Liel Maruani lielluse@post.bgu.ac.il.

1. MULTIPLE QUBITS AND QUANTUM CIRCUITS

For each state $|x_0, x_1\rangle$ for $x_i \in \{0, 1\}$, apply the following circuit:



Repeat the calculation with the following circuit:



Conclude that these two unitary transformations are equal, and that the intuition that the C-NOT gate does not change the control qubit is *false*, when we work in the $\{|+\rangle, |-\rangle\}$ basis.

2. DO WE REALLY NEED COMPLEX NUMBERS?

Prove that we can assume w.l.o.g. that all amplitudes in a quantum computation are real numbers!

Guidance: Show that any quantum circuit on n qubits that uses t two-qubit gates with complex coefficients can be simulated exactly by a quantum circuit on $n+1$ qubits, that uses at most t three-qubit gates with real coefficients. Show that the probability of acceptance remains the same in your construction.

Hint: Remember that we assume that the input for a quantum circuit is in the standard basis (so, in the base of our induction, we know that the state has only real amplitudes). Use the extra qubit to decide whether the original state is in the Imaginary or Real part, and adapt each gate to a gate with one more qubit, such that the new gate works in an equivalent way to the old one. More formally, suppose that at some step of the original circuit the state at that step is $\sum_x (a_x + ib_x)|x\rangle$, the state at that step in the new circuit should be $\sum_x a_x|0x\rangle + b_x|1x\rangle$.

3. RECONSTRUCTING A MATRIX

Let A be a binary $n \times n$ matrix. For $x, y \in \{0, 1\}^n$, let:

$$f(x, y) = x^T A y \mod 2 = \left(\sum_{i,j} A_{i,j} x_i y_j \right) \mod 2.$$

Convince yourself (there is no need to submit this argument) that there exists a simple classical algorithm that reconstructs A using n^2 queries to f , and that there is no algorithm that can do that with fewer classical queries.

Find a quantum algorithm that reconstructs the matrix A using $O(n)$ queries to the corresponding quantum phase oracle. Recall that the quantum phase oracle for f acts as follows: For every $x, y \in \{0, 1\}^n$,

$$V_f|x, y\rangle = (-1)^{(\sum_{i,j} A_{i,j} x_i y_j) \bmod 2} |x, y\rangle.$$

4. GROVER'S ALGORITHM WITH AN UNKNOWN NUMBER OF MARKED ITEMS

You are given access to an oracle with K marked items, and K is unknown. Design an algorithm that finds a marked item using $O(\sqrt{N/K})$ queries, with probability more than $\frac{1}{2}$.¹

Guidance:

- (1) Show that when K is known, there is an algorithm that finds the marked element with probability close to 1, with running time $O(\sqrt{\frac{N}{K}})$.
- (2) Show that if K lies between 2^j and 2^{j+1} for a known j , there is an algorithm that finds a marked element with probability at least $\frac{1}{2}$, with the same running time.
- (3) Iterate over the possible values of j in a certain order, which gives a total running time of $O(\sqrt{\frac{N}{K}})$ with probability $\frac{1}{2}$.

¹Your algorithm does not need to run in expected running time $O(\sqrt{N/K})$. Such an algorithm exists though it is slightly harder than the one you need to find in this exercise.